SUMMARY

The elements belonging to groups 1 and 2 of the modern periodic table are called s-block elements. They are called so because the valence electron occupies the s orbitals. The group 1 elements have a general outer electronic configuration ns^1 and are called alkali metals. The group 2 elements have a general outer electronic configuration ns^2 and these are called alkaline earth metals as they are found in earth's crust and their oxides and hydroxides are alkaline in nature. Elements belonging to group 1 and 2 are highly reactive and forms M^+ and M^{2+} cations respectively. Their physical and chemical properties of both groups show a regular trend as we move down the group. The atomic and ionic radii increase as we move down the group while their ionisation enthalpies decrease.

The first element in each of these groups, lithium in Group 1 and beryllium in Group 2 shows some difference in behaviour with the elements in rest of their groups and show similarities in properties to the second member of the next group. This behaviour is known as the 'diagonal relationship' in the periodic table.

The alkali metals are soft and silvery white in colour with low melting points. They are highly reactive. The compounds of alkali metals are predominantly ionic. They form metal hydrides and halides with hydrogen and halogens respectively. Their oxides and hydroxides are soluble in water forming strong alkalies. Important compounds of sodium include sodium carbonate, sodium chloride, sodium hydroxide and sodium hydrogen carbonate. Sodium hydroxide is manufactured by Castner-Kellner process and sodium carbonate by Solvay process.

The chemistry of alkaline earth metals is similar to alkali metals. However, we observe some differences because of their reduced atomic and ionic sizes and increased cationic charges. Their oxides and hydroxides are less basic than the alkali metal oxides and hydroxides. They also form hydrides and halides with hydrogen and halogens respectively. Industrially important compounds of calcium include calcium oxide (lime), calcium hydroxide (slaked lime), calcium sulphate hemihydrate (Plaster of Paris), calcium carbonate (limestone) and cement. Portland cement is an important constructional material. It is manufactured by heating a pulverised mixture of limestone and clay in a rotary kiln. The clinker thus obtained is mixed with some gypsum (2-3%) to give a fine powder of cement. All these substances find variety of uses in different areas.

Monovalent sodium and potassium ions and divalent magnesium and calcium ions are found in large proportions in biological fluids. These ions perform important biological functions such as maintenance of ion balance and nerve impulse conduction.

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